

Math 142, S12

Quiz 10

Section:

Name: Answer

SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No calculator is allowed in the quiz and exam.

Problem 1. (6 points) Test the following series for convergence or divergence.

(a) $\sum_{n=1}^{+\infty} \frac{n}{3^n}$ (use Ratio Test)

$$a_n = \frac{n}{3^n}$$

$$\left| \frac{a_{n+1}}{a_n} \right| = \left| \frac{\frac{n+1}{3^{n+1}}}{\frac{n}{3^n}} \right| = \left| \frac{n+1}{3^{n+1}} \cdot \frac{3^n}{n} \right|$$

$$= \left| \frac{n+1}{3n} \right|$$

$$= \left| \frac{1 + \frac{1}{n}}{3} \right| \rightarrow \frac{1}{3} \text{ as } n \rightarrow +\infty$$

$$\frac{1}{3} < 1$$

$$\Rightarrow \sum_{n=1}^{+\infty} \frac{n}{3^n} \text{ is convergent.}$$

(b) $\sum_{n=2}^{+\infty} \frac{1}{n\sqrt{\ln n}}$ (use Integration Test)

$$f(x) = \frac{1}{x\sqrt{\ln x}} \text{ decreasing function on } [2, +\infty)$$

$$\int_2^{+\infty} f(x) dx = \int_2^{+\infty} \frac{1}{x\sqrt{\ln x}} dx$$

$$\stackrel{u=\ln x}{=} \int_{\ln 2}^{+\infty} \frac{1}{\sqrt{u}} du$$

$$= \lim_{t \rightarrow +\infty} (2\sqrt{u}) \Big|_{\ln 2}^t$$

$$= \lim_{t \rightarrow +\infty} (2\sqrt{t} - 2\sqrt{\ln 2})$$

$$= +\infty$$

$$\Rightarrow \sum_{n=2}^{+\infty} \frac{1}{n\sqrt{\ln n}} \text{ is divergent.}$$

Problem 2. (4 points) Find the radius of convergence of the power series $\sum_{n=0}^{+\infty} \frac{n x^n}{2^n}$.

$$a_n = \frac{n x^n}{2^n}$$

$$\left| \frac{a_{n+1}}{a_n} \right| = \left| \frac{\frac{(n+1)x^{n+1}}{2^{n+1}}}{\frac{n x^n}{2^n}} \right| = \left| \frac{(n+1)x^{n+1}}{2^{n+1}} \cdot \frac{2^n}{n x^n} \right|$$

$$= \left| \frac{n+1}{n} \cdot \frac{x}{2} \right| = \left| \frac{1 + \frac{1}{n}}{1} \cdot \frac{x}{2} \right| \rightarrow \left| \frac{x}{2} \right| \text{ as } n \rightarrow +\infty$$

$$\left| \frac{x}{2} \right| < 1 \Rightarrow |x| < 2 \Rightarrow R = 2.$$

If we want to find the interval of convergence, then first $\sum_{n=0}^{+\infty} \frac{n x^n}{2^n}$ is convergent in $(-2, 2)$.

For $x = -2$, we get $\sum_{n=0}^{+\infty} \frac{n(-2)^n}{2^n} = \sum_{n=0}^{+\infty} (-1)^n n \rightarrow \text{divergent}$

For $x = 2$, we get $\sum_{n=0}^{+\infty} \frac{n 2^n}{2^n} = \sum_{n=0}^{+\infty} n \rightarrow \text{divergent}$.

Thus, the interval of convergence is $(-2, 2)$.